

ECON403/503—Midterm I (sketch)

Fall 2021

Mapachonzzzito

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v1.1

1 Problem 1

1.1 Type I exam

(40 points) Consider a decision maker (DM) that has preference over lotteries, with $Z \subseteq \mathbb{R}_{++}$ being the set of monetary prizes¹. Let $v : Z \rightarrow \mathbb{R}$; then the DM assigns the number $v(z)$ to each amount of money $z \in Z$. The function v is continuous and increasing, and strictly positive. Define $U : \mathcal{L}(Z) \rightarrow \mathbb{R}$ by

$$U(p) = \alpha \max\{v(z) \mid p(z) > 0\} + (1 - \alpha) \min\{v(z) \mid p(z) > 0\},$$

where $0 \leq \alpha \leq 1$. Let $\succeq$ denote the preference relation on $\mathcal{L}(Z)$ that is represented by U .

1. **(20 points)** Does the DM's preference over lotteries $\succeq$ admit a Expected Utility representation?
2. **(20 points)** The notion of *risk aversion* is independent of Expected Utility Theory, so irrespective of your answer to the previous item, we can define aversion as follows.² A DM is risk averse if she always prefers the expected value of any lottery with certainty to the lottery itself, that is, for any $p \in \mathcal{L}(Z)$, $[E(p)] \succeq p$. Characterize decision makers with this type of preference that are risk averse for any α and v .

1.2 Type II exam

(40 points) Consider a decision maker (DM) that has preferences over lotteries, with $Z \subseteq \mathbb{R}_{++}$ being the set of monetary prizes³. Let $v : Z \rightarrow \mathbb{R}$; then the DM assigns the number $v(z)$ to each amount of money $z \in Z$. The function v is continuous and increasing, and strictly positive. Define $U : \mathcal{L}(Z) \rightarrow \mathbb{R}$ by

$$U(p) = \alpha \max \left\{ \frac{p(z)}{v(z)} \mid z \in Z \right\},$$

where α is any real number different from zero. Let $\succeq$ denote the preference relation on $\mathcal{L}(Z)$ that is represented by U .

1. **(20 points)** Does the DM's preferences over lotteries $\succeq$ admit a Expected Utility representation?
2. **(20 points)** The notion of *risk aversion* is independent of Expected Utility Theory, so irrespective of your answer to the previous item, we can define aversion as follows.⁴ A DM is risk averse if she always prefers the expected value of any lottery with certainty to the lottery itself, that is, for any $p \in \mathcal{L}(Z)$, $[E(p)] \succeq p$. Characterize decision makers with this type of preference that are risk averse for any α and v .

¹The notation $\mathbb{R}_{++}$ denotes the strictly positive part of the real line.

²Notation: $E(p)$ denotes the expected value of the lottery p and $[y]$ denotes the degenerate lottery that awards y with probability one.

³The notation $\mathbb{R}_{++}$ denotes the strictly positive part of the real line.

⁴Notation: $E(p)$ denotes the expected value of the lottery p and $[y]$ denotes the degenerate lottery that awards y with probability one.

2 Problem 2

2.1 Type I exam

(20 points) A price-taking firm, i.e., operating under perfect competition, seeks to maximize profits. The firm produces a good y , which it sells at price p , and faces costs of production given by $c(y, w_1, w_2) = y^2(w_1 + w_2)$, where w_1 and w_2 are the prices of two inputs in the production of y . However, the firm faces uncertainty regarding the price p : with probability λ the price is p_h and with probability $1 - \lambda$ the price is p_l , with $p_h > p_l$.

1. **(10 points)** State the firm's problem and solve for the supply function of the firm, i.e., the optimal value of y as a function of the prices and λ . Be as rigorous as you can.
2. **(10 points)** Now suppose that p is a continuous random variable that can take values on the interval $[p_l, p_h]$ and is distributed according to the probability density function f . State the firm's problem and solve for the supply function. Be as rigorous as you can.

2.2 Type II exam

(20 points) A price-taking firm, i.e., operating under perfect competition, seeks to maximize profits. The firm produces a good y , which it sells at price p , and faces costs of production given by $c(y, w_1, w_2) = y^2(w_1 + w_2)$, where w_1 and w_2 are the prices of two inputs in the production of y . However, the firm faces uncertainty regarding the input price w_1 : with probability θ the price is $w_{1,h}$ and with probability $1 - \theta$ the price is $w_{1,l}$, with $w_{1,h} > w_{1,l}$.

1. **(10 points)** State the firm's problem and solve for the supply function of the firm, i.e., the optimal value of y as a function of the prices and θ . Be as rigorous as you can.
2. **(10 points)** Now suppose that w_1 is a continuous random variable that can take values on the interval $[w_{1,l}, w_{1,h}]$ and is distributed according to the probability density function g . State the firm's problem and solve for the supply function. Be as rigorous as you can.

3 Problem 3

3.1 Type I Exam

(40 points) Consider a situation where all individuals face an idiosyncratic risk⁵ to lose 100 with probability p . Suppose there are N individuals. The individuals decide to create a mutual agreement, where the aggregate loss in the pool will be equally split among its members.

1. **(10 points)** Describe explicitly in terms of lotteries the residual loss⁶ of each member of the pool when $N = 2$ and $N = 3$.
2. **(10 points)** Give a formal proof that an increase from $N = 2$ to $N = 3$ generates a reduction in risk borne by each member in the sense of Second Order Stochastic Dominance.
3. **(20 points)** Generalize the result for the case of N and $N + 1$, where N is an arbitrary but finite number of individuals.

3.2 Type II Exam

(40 points) Suppose that an individual can buy shares of K assets, $X_1, \dots, X_K$ whose return next period (per dollar invested) is described by identical and independent distributions. The individual has wealth w to invest and has a strictly concave and strictly increasing Bernoulli utility function u over wealth. Let $\alpha = (\alpha_1, \dots, \alpha_K)$ denote the portfolio of the individual, i.e., $0 \leq \alpha_i \leq 1$ is the share on asset i , with $\sum_{i=1}^K \alpha_i = 1$.

1. **(10 points)** State, *but do not solve*, the problem that the individual needs to solve to choose the optimal portfolio.

⁵Idiosyncratic: adjective denoting peculiar or individual. Thus, idiosyncratic risk means that each individual faces this risk and that the probability of one individual facing this risk is independent of the others.

⁶If an individual with wealth w loses an amount L , the residual loss is $-L$.

2. **(10 points)** If $K = 2$, show that no matter what u is, the individual should always choose the portfolio $(\alpha_1, \alpha_2) = (1/2, 1/2)$.
3. **(20 points)** Let K be a finite but arbitrary natural number. Generalize the result above: show that no matter what u is, the individual should always choose the portfolio $(1/K, \dots, 1/K)$.